Name: ___________________________

Registration: _____________________

(10 pts.) Given the following partial module for a 128Kx32 RAM chip, fill in the blanks to complete the module.

```
module RAM (adr, CS, RW, di, do);

    input CS, RW;
    input [      ] adr;
    input [      ] di;
    output [      ] do;
    reg [      ] d_out;
    reg [      ] Mem1 [        ];

    assign do = (CS && RW)?d_out:_______;

    always @(adr or di or CS or RW)
         if (CS && !RW)
              Mem1 [      ] = _______;
    always @(adr or CS or RW)
         if (CS && RW)
              d_out = Mem1 [      ];

    initial
         $readmemh ("memory1.dat", Mem1);

endmodule
```